

Orthogonality, Projection, and Least Square Solution

Orthogonality and Orthonormal Sets

1. Orthogonality

- **Definition:** Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are *orthogonal* if their dot product is zero: $\mathbf{u} \cdot \mathbf{v} = 0$.
- **Geometric Interpretation:** Non-zero orthogonal vectors are perpendicular to each other. If either vector is the zero vector, they are considered orthogonal by definition.
- **Theorem:** If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, then for any scalars $s, t \in \mathbb{R}$, $s\mathbf{u}$ and $t\mathbf{v}$ are also orthogonal.
 - *Proof:* $(s\mathbf{u}) \cdot (t\mathbf{v}) = st(\mathbf{u} \cdot \mathbf{v}) = st(0) = 0$.

2. Orthogonal and Orthonormal Sets

- **Orthogonal Set Definition:** A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ in $\mathbb{R}^n$ is *orthogonal* if every pair of distinct vectors in S is orthogonal: $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for all $i \neq j$.
- **Orthonormal Set Definition:** A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is *orthonormal* if it is orthogonal, and each vector is a unit vector (i.e., has a norm of 1):
 - $\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$
- **Key Properties:**
 - An orthogonal set of *nonzero* vectors is linearly independent.
 - An orthonormal set is always linearly independent.

3. Orthogonal to a Subspace

- **Definition:** A vector $\mathbf{n}$ is *orthogonal* to a subspace V of $\mathbb{R}^n$ if $\mathbf{n}$ is orthogonal to every vector in V . This is denoted as $\mathbf{n} \perp V$.
- **Characterization:** To show $\mathbf{n} \perp V$, it is sufficient to show that $\mathbf{n}$ is orthogonal to every vector in a spanning set for V .
- **Theorem:** If V is a subspace of $\mathbb{R}^n$ and $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a spanning set for V , then a vector $\mathbf{w}$ is orthogonal to V if and only if $\mathbf{w} \cdot \mathbf{u}_i = 0$ for all $i = 1, \dots, k$.

4. Orthogonal Complement

- **Definition:** The *orthogonal complement* of a subspace V in $\mathbb{R}^n$, denoted by $V^\perp$, is the set of all vectors in $\mathbb{R}^n$ that are orthogonal to every vector in V :
 - $V^\perp = \{\mathbf{w} \in \mathbb{R}^n \mid \mathbf{w} \cdot \mathbf{v} = 0 \text{ for all } \mathbf{v} \in V\}$.
- **Key Properties:**
 - $V^\perp$ is itself a subspace of $\mathbb{R}^n$.
 - If V is a subspace spanned by $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$, then $V^\perp = \text{Null}(A^T)$, where A is the matrix whose columns are the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$.

Orthogonal and Orthonormal Bases & Orthogonal Projections

1. Orthogonal and Orthonormal Bases

- **Definition:**
 - An *orthogonal basis* for a subspace V is a basis that is also an orthogonal set.
 - An *orthonormal basis* for a subspace V is a basis that is also an orthonormal set.

- **Theorem:**

Suppose $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$. Then S is linearly independent.

- **Proof:**

Let $c_1, c_2, \dots, c_k$ be coefficients such that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k = \mathbf{0}.$$

For each $i = 1, \dots, k$,

$$0 = \mathbf{u}_i \cdot \mathbf{0} = \mathbf{u}_i \cdot (c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \cdots + c_k \mathbf{u}_k) = c_1 \mathbf{u}_i \cdot \mathbf{u}_1 + \cdots + c_i \mathbf{u}_i \cdot \mathbf{u}_i + \cdots + c_k \mathbf{u}_i \cdot \mathbf{u}_k = c_i \|\mathbf{u}_i\|^2,$$

since $\mathbf{u}_i$ orthogonal to $\mathbf{u}_j$ for each $i \neq j$.

Because $\mathbf{u}_i$ are nonzero, thus $\mathbf{u}_i \neq \mathbf{0}$ and $\|\mathbf{u}_i\|^2 > 0$.

Hence, $c_i = 0$ and S linearly independent.

2. Coordinates Relative to an Orthogonal Basis

- Theorem:** Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be an orthogonal basis for a subspace V of $\mathbb{R}^n$. Then, for any vector $\mathbf{v} \in V$:

$$\begin{aligned} \bullet \quad \mathbf{v} &= \frac{\mathbf{v} \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 + \frac{\mathbf{v} \cdot \mathbf{u}_2}{\|\mathbf{u}_2\|^2} \mathbf{u}_2 + \cdots + \frac{\mathbf{v} \cdot \mathbf{u}_k}{\|\mathbf{u}_k\|^2} \mathbf{u}_k \\ \bullet \quad [\mathbf{v}]_S &= \begin{pmatrix} \frac{\mathbf{v} \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2} \\ \frac{\mathbf{v} \cdot \mathbf{u}_2}{\|\mathbf{u}_2\|^2} \\ \vdots \\ \frac{\mathbf{v} \cdot \mathbf{u}_k}{\|\mathbf{u}_k\|^2} \end{pmatrix} \end{aligned}$$

- For Orthonormal Basis:**

If S is orthonormal, then the coordinates simplify to:

$$\begin{aligned} \bullet \quad \mathbf{v} &= (\mathbf{v} \cdot \mathbf{u}_1) \mathbf{u}_1 + (\mathbf{v} \cdot \mathbf{u}_2) \mathbf{u}_2 + \cdots + (\mathbf{v} \cdot \mathbf{u}_k) \mathbf{u}_k \\ \bullet \quad [\mathbf{v}]_S &= \begin{pmatrix} \mathbf{v} \cdot \mathbf{u}_1 \\ \mathbf{v} \cdot \mathbf{u}_2 \\ \vdots \\ \mathbf{v} \cdot \mathbf{u}_k \end{pmatrix} \end{aligned}$$

- Standard Basis:**

If $E = \{e_1, e_2, \dots, e_n\}$ is an orthonormal basis for $\mathbb{R}^n$, then for $\mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$$

The components are given by $e_i \cdot \mathbf{v} = v_i$.

Thus

$$[\mathbf{v}]_E = \mathbf{v}.$$

3. Orthogonal Projection

- Theorem (Orthogonal Projection Theorem):** Let V be a subspace of $\mathbb{R}^n$. Every vector $\mathbf{w}$ in $\mathbb{R}^n$ can be uniquely decomposed as:
 - $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$, where $\mathbf{w}_p \in V$, $\mathbf{w}_n \perp V$.
 - $\mathbf{w}_p$ is called the *orthogonal projection* of $\mathbf{w}$ onto V .
- Formula for Orthogonal Projection:** If $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthogonal basis for V , then:
 - $\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 + \frac{\mathbf{w} \cdot \mathbf{u}_2}{\|\mathbf{u}_2\|^2} \mathbf{u}_2 + \cdots + \frac{\mathbf{w} \cdot \mathbf{u}_k}{\|\mathbf{u}_k\|^2} \mathbf{u}_k$
 - $\mathbf{w}_n = \mathbf{w} - \mathbf{w}_p$
- Best Approximation Theorem:** The orthogonal projection $\mathbf{w}_p$ is the vector in V that is closest to $\mathbf{w}$. That is, for any $\mathbf{v} \in V$:
- $\|\mathbf{w} - \mathbf{w}_p\| \leq \|\mathbf{w} - \mathbf{v}\|$

4. Gram-Schmidt Process

- Purpose:** To construct an orthogonal (or orthonormal) basis for a subspace V , given any basis for V .
- Algorithm:** Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for V . Define:
 - $\mathbf{v}_1 = \mathbf{u}_1$
 - $\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \mathbf{v}_1$
 - $\mathbf{v}_3 = \mathbf{u}_3 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\mathbf{u}_3 \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2} \mathbf{v}_2$
 - $\dots$
 - $\mathbf{v}_k = \mathbf{u}_k - \sum_{i=1}^{k-1} \frac{\mathbf{u}_k \cdot \mathbf{v}_i}{\|\mathbf{v}_i\|^2} \mathbf{v}_i$
 - Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an orthogonal basis for V .

- **To obtain an orthonormal basis**, normalize each vector $\mathbf{v}_i$: $\mathbf{q}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$.
- If the Gram-Schmidt process results in a zero vector, that vector is discarded, and the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots$ were not linearly independent.

5. Orthogonal Matrices

- **Definition:**
A matrix A of size $n \times n$ is *orthogonal* if $A^T = A^{-1}$. Equivalently $A^T A = I$ and $AA^T = I$, where I is the identity matrix.
- **Theorem:**
Suppose A is a square matrix of order n . Then the following statements are equivalent.
 - A is an orthogonal matrix.
 - The columns of A form an orthonormal basis for $\mathbb{R}^n$.
 - The rows of A form an orthonormal basis for $\mathbb{R}^n$.

Week 9: Least Square Approximation

1. Least Square Approximation

- **Motivation:** What if $A\mathbf{x} = \mathbf{b}$ has no solution?
- **Definition:** Given an $m \times n$ matrix A and a vector $\mathbf{b}$ in $\mathbb{R}^m$, a vector $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ is a *least-squares solution* of $A\mathbf{x} = \mathbf{b}$ if $\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$ for all $\mathbf{x}$ in $\mathbb{R}^n$. Geometrically, we're finding $A\hat{\mathbf{x}}$ in $\text{Col}(A)$ that's closest to $\mathbf{b}$.
- **Normal Equations:** $\hat{\mathbf{x}}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if $\hat{\mathbf{x}}$ satisfies the *normal equations*:
 - $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$
- **Uniqueness of Solution:**
 - If the columns of A are linearly independent, then A has full rank, $A^T A$ is invertible, and there is a unique least squares solution $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$.
- For any choice of a least square solution $\mathbf{u}$, the projection $A\mathbf{u}$ is unique.

2. Orthogonal Projection (Revisited)

- **Connection to Least Squares:** If $\mathbf{w}_p$ is the orthogonal projection of $\mathbf{w}$ onto $\text{Col}(A)$, then $A\hat{\mathbf{x}} = \mathbf{w}_p$.
- **Least Squares Solution for Orthogonal Projection:** If we have a least squares solution to $A\mathbf{x} = \mathbf{w}$, say $\hat{\mathbf{x}}$, then $A\hat{\mathbf{x}} = A(A^T A)^{-1} A^T \mathbf{w} = \mathbf{w}_p$.
- **Theorem:**
If A is a $m \times n$ matrix such that the columns of A are linearly independent, for a vector $\mathbf{u}$ in $\mathbb{R}^n$:

$$A(A^T A)^{-1} A^T \mathbf{u} = \text{orthogonal projection of } \mathbf{u} \text{ onto } \text{Col}(A)$$

$$\mathbf{w}_p = A(A^T A)^{-1} A^T \mathbf{w}.$$

Important notes about subspaces V and their Nullspaces

- Column Space: A subspace of $\mathbb{R}^m$ (where A is a $m \times n$ matrix)
- Row Space: A subspace of $\mathbb{R}^n$ (where A is a $m \times n$ matrix)
- Nullspace: $A\mathbf{x} = \mathbf{0}$ and the solutions to this lies in $\mathbb{R}^n$
- Orthogonal Complement: A subspace

Important Note: if V is subspace, and $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$

Then the $V^\perp = \text{Null}(A^T)$,

where $A = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k)$

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- **Gram-Schmidt process and Q, R decomposition:**

```
function [Q,R] = gram_schmidt(A)
[m,n] = size(A);
Q = zeros(m,n);
```

```

R = zeros(n,n);

for j = 1:n
    v = A(:,j);
    for i = 1:j-1
        R(i,j) = Q(:,i)' * A(:,j);
        v = v - R(i,j) * Q(:,i);
    end
    R(j,j) = norm(v);
    Q(:,j) = v / R(j,j);
end
end

```

- Check orthogonality and orthonormality

```

function [orthogonality, orthonormality] = check_properties(matrix)
% Check Orthogonality
G = matrix'*matrix;
off_diagonal_elements = G - diag(diag(G));
orthogonality = norm(off_diagonal_elements, 'fro') < 1e-10; % Frobenius norm
% Check Orthonormality
diagonals = diag(G);
orthonormality = orthogonality && all(abs(diagonals - 1) < 1e-10);
end

```

- Orthogonal projection

```

% Given w and orthogonal matrix A
wp = A*((A'*A)\(A'*w)); % This works when columns of A are linearly independent (a basis)
% General projection
wp = A*pinv(A'*A)*A'*w; % pinv is a pseudo-inverse

```